

PRESENTATION_GUIDE

COMPLETE PROJECT EXPLANATION SUMMARY

PROJECT OVERVIEW

The **Mental Health Risk Prediction System** is an AI-powered tool that uses machine learning to assess mental health risk levels (Low/Medium/High) based on 13 demographic and clinical factors. Built with Python, scikit-learn, and Gradio, it achieves 100% accuracy in predicting mental health risk levels.

TECHNICAL ARCHITECTURE

- **Algorithm:** Decision Tree Classifier
- **Features Analyzed:** 13 variables (age, gender, employment, stress, depression, anxiety, etc.)
- **Interface:** Web-based Gradio application
- **Performance:** 100% accuracy across all metrics
- **Languages:** Python, HTML/CSS for interface

PERFORMANCE METRICS

- **Accuracy:** 1.0000 (100%)
- **Precision:** 1.0000 (Perfect prediction quality)
- **Recall:** 1.0000 (Perfect case identification)
- **F1-Score:** 1.0000 (Perfect balance)
- **MSE:** 0.0000 (No prediction errors)
- **MAE:** 0.0000 (No absolute errors)
- **R² Score:** 1.0000 (Perfect variance explanation)

KEY FEATURES

1. **Real-time Prediction:** Instant risk assessment
2. **Explanation Cards:** 7 detailed documentation tabs
3. **Sample Cases:** Predefined test scenarios
4. **Copy Functionality:** Easy result sharing
5. **Comprehensive Metrics:** All performance indicators
6. **User-Friendly Interface:** Intuitive web navigation
7. **Detailed Explanations:** Reasoning for each prediction



PROBLEM SOLVED

- **Mental Health Crisis:** 1 in 5 adults experience mental illness annually
- **Delayed Treatment:** Average 11-year gap between onset and treatment
- **Lack of Screening Tools:** Insufficient early detection methods
- **Stigma Barriers:** 60% deterred from seeking help due to stigma



SOLUTION PROVIDED

- **Early Detection:** Identifies at-risk individuals before crisis
- **Accessibility:** Web-based platform available 24/7
- **Professional Support:** Aids healthcare providers in assessment
- **Individual Empowerment:** Self-assessment tool for users
- **Resource Optimization:** Helps prioritize high-risk cases



HOW TO PRESENT

Opening

- Introduce yourself and team
- State the project title clearly
- Briefly mention the purpose

Main Presentation

- Follow the 7-slide structure
- Emphasize the 100% accuracy
- Highlight the 13 factors analyzed
- Explain the real-world impact

Demo (if possible)

- Show the live application
- Run a sample prediction
- Demonstrate the explanation cards
- Show the metrics tab

Closing

- Summarize achievements
- Mention future potential
- End with inspiring message



PITCH SCRIPT

Introduction: "Today I'm presenting our Mental Health Risk Prediction System, an AI-powered tool that addresses the critical gap in mental health screening."

Problem: "With 1 in 5 adults experiencing mental illness and an average 11-year delay in treatment, early detection is crucial."

Solution: "Our system uses machine learning to provide instant, accurate risk assessments with detailed explanations."

Results: "We achieved perfect 100% accuracy across all metrics while maintaining user-friendly design."

Impact: "This tool can transform mental health screening by making it accessible, accurate, and immediate."

Conclusion: "We're ready to revolutionize mental health screening through artificial intelligence."



TIPS FOR PRESENTATION

- **Practice:** Know your content well
- **Confidence:** Speak clearly and maintain eye contact
- **Engagement:** Ask rhetorical questions
- **Timing:** Allocate ~3-5 minutes per slide
- **Q&A:** Be prepared for technical and ethical questions
- **Demo:** Have backup plan if live demo fails

This project demonstrates the power of AI in healthcare, combining technical excellence with meaningful social impact.